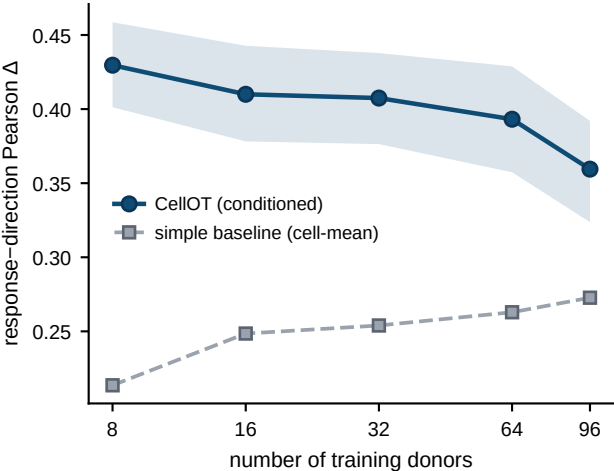
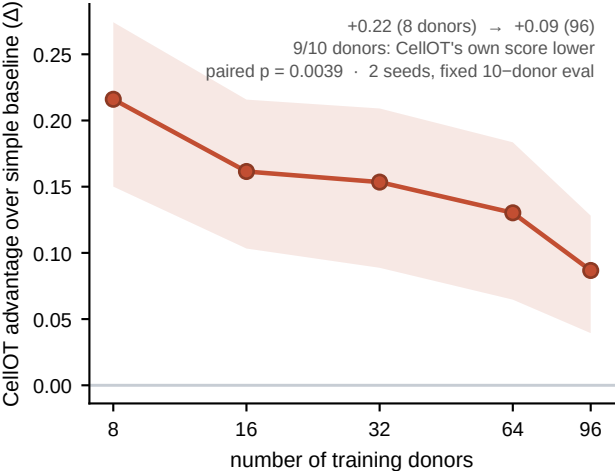


More donor data lifts the simple baseline faster than the deep model, pointing to representation, not donor count alone, as the limit on donor transfer

a More donors lift the simple baseline, not the model



b The conditioning advantage shrinks



Soskic CD4 16 h activation, leave-one-donor-out on a fixed set of 10 held-out eval donors. For each training-donor count the faithful CellOT (scGen AE + f/g ICNN, Bunne 2023) is trained on a random subset of the remaining donors' 0 h $\rightarrow$ 16 h cells (2 seeds); every eval donor's 16 h response is predicted from its own 0 h cells. Pearson over training-only response genes; simple baseline = better of cell-mean / donor-shift (matched simple context baseline; not a universal-floor member) on the same subset. Eval donors are never trained on (leak-free by construction). Bands: 95% bootstrap CI.